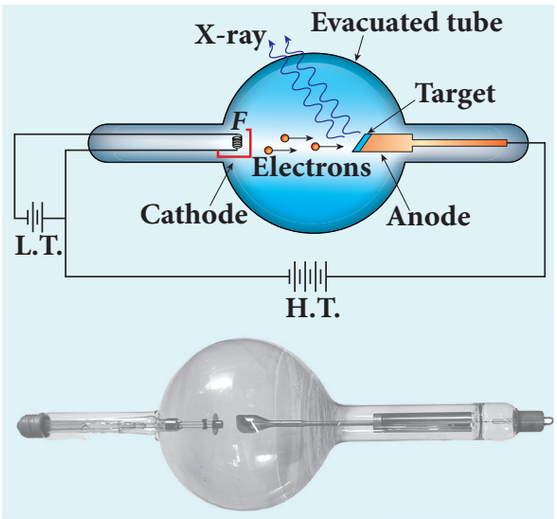



**Figure 8.20** Production of x-rays

When high-speed electrons strike the target, they are decelerated suddenly and lose their kinetic energy. As a result, x-ray photons are produced. Since most of the kinetic energy of the bombarding electrons gets converted into heat, targets made of high-melting-point metals and a cooling system are usually employed.

### X-ray spectra

X-rays are produced when fast moving electrons strike the metal target. The intensity of the x-rays when plotted against its wavelength gives a curve called **x-ray spectrum** (Figure 8.21(a) and (b)). X-ray spectra consist of two parts: a continuous spectrum and a series of peaks superimposed on it.

The **continuous spectrum** consists of radiations of all possible wavelengths with a certain minimum wavelength $\lambda_0$ which depends on the voltage across the electrodes. The peaks are characteristics of the material of the target and hence it is called **characteristic spectrum**. Figure 8.21(a) depicts the x-ray spectra of tungsten at various accelerating voltages and Figure 8.21(b) shows the x-ray spectra of tungsten and molybdenum at a particular accelerating voltage.

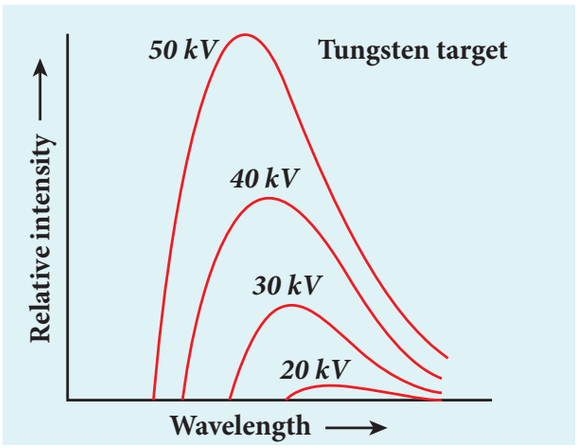



**Figure 8.21** (a) X-ray spectra of tungsten at various accelerating potentials

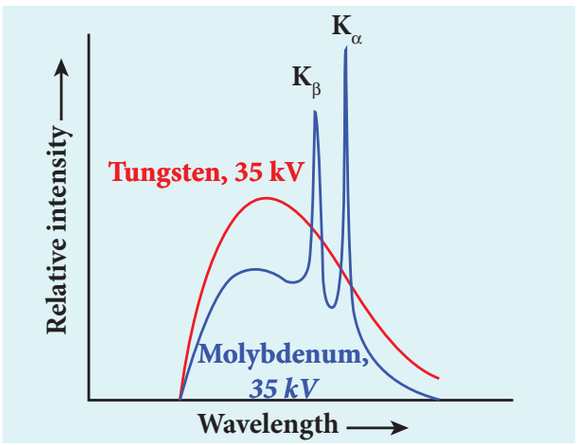



**Figure 8.21** (b) X-ray spectra of tungsten and molybdenum at 35 *kV* accelerating potential

Though classical electromagnetic theory suggests the emission of radiations from accelerating electrons, it could not explain two features exhibited by x-ray spectra. These features are given below.

(i) For a given accelerating voltage, the lower limit for the wavelength of continuous x-ray spectra is same for all targets. This minimum wavelength is called cut-off wavelength.

(ii) The intensity of x-rays is significantly increased at certain well-defined wavelengths as shown in the case of characteristic x-ray spectra for molybdenum (Figure 8.21(b)).



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